RÉPUBLIQUE ALGÉRIENNE DÉMOCRATIQUE ET POPULAIRE
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In : **Software Engineering**

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CHAPTER 1
INTRODUCTION

1.1 Purpose

This document provides a comprehensive overview of the platform designed to facilitate interaction and management within an educational institution. It outlines the system's objectives, functionalities, and requirements to guide development and implementation.

1.2 Product scope

The platform will connect students, teachers, and administration. It provides students access to their grades, course information, and updates. Additionally, it offers a chatbot for quick answers to common queries and a functionality for students to appeal their grades. Teachers can manage marks, deliberate on appeals, and the system will automatically calculate final grades based on predefined criteria. The platform will also integrate a feature for students to use their ID cards digitally.

1.3 Glossery

- **SRS (Software Requirements Specification)**: A document detailing the functional and non-functional requirements of a software system.

- **DB (Database)**: A collection of electronically organized data used by the system to manage user information, courses, and grades.

- **CRUD (Create, Read, Update, Delete)**: The basic operations used to manage data in a database.

- **API (Application Programming Interface)**: A set of functions allowing different applications to communicate with each other.
- **AI (Artificial Intelligence)**: Technology used to develop a chatbot capable of answering student questions and assisting teachers in administrative tasks.
- **ID (Identification)**: A digital identification card used to authenticate users on the platform.
- **User Management Module**: The component of the system that handles authentication and profiles for students, teachers, and administrators.
- **Course and Grade Management**: A feature allowing teachers to manage courses and enter student grades, while students can view their courses and results.
- **Chatbot**: An AI-powered virtual assistant integrated into the system to respond to user queries and assist teachers with tasks.
- **Grade Calculation Engine**: An automated system that calculates students' final grades based on predefined rules set by the institution.
- **Appeal and Deliberation Management**: A feature allowing students to appeal their grades and teachers to deliberate and process these appeals.
- **Digital Identification Card**: A digital card that allows students to securely access platform services.
- **Data Security**: Measures taken to protect sensitive user information, such as grades and credentials.
- **Accessibility**: Compliance with standards that enable all users, including those with disabilities, to access the platform easily.

- **Availability**: The system's ability to operate without interruption, ensuring continuous access to the platform for all users.

- **GDPR Compliance (General Data Protection Regulation)**: Data protection standards that the platform must adhere to, ensuring the privacy and security of users' personal information.

1.4 References

-Educational institution policies and grading rules.
-Technical references for technologies and frameworks used (e.g., databases, APIs).

1.5 Overview

This SRS will detail the functionalities, system architecture, external interfaces, and both functional and non-functional requirements of the platform.

CHAPTER 2

GENERAL DESCRIPTION

2.1 Product perspective

The platform will be a web-based application integrating several services to create an efficient academic management system. The key components include:

2.1.1 User Management Module

Handles student, teacher, and administration profiles and authentication using digital ID cards.

2.1.2 Course and Grade Management

Teachers input grades, students view their results and course information, and the platform allows for grade appeals and updates.

2.1.3 Chatbot Module

An AI-powered assistant to respond to student queries, reducing the workload on teachers and administrative staff.

2.1.4 Grade Calculation Engine

An automated system that calculates final grades based on rules set by the institution.

2.2 Product function

2.2.1 User Authentication and Authorization

Secure login for students, teachers, and administration, supporting the use of digital ID cards.

2.2.2 Course Management

Teachers can create, update, and manage courses and course-related materials.

2.2.3 Grade Management

Teachers can input grades, the system automatically calculates final grades, and students can appeal grades.

2.2.4 Chatbot Assistance

An AI chatbot that answers frequently asked questions, guides students, and provides assistance to teachers.

2.2.5 Appeal and Deliberation System

Students can submit grade appeals, and teachers can review and deliberate on these appeals.

2.2.6 Digital ID Card Integration

The platform supports using digital versions of student ID cards for authentication and accessing services.

2.2.7 Reporting and Analytics

Provides reports on student performance and other relevant metrics for teachers and administrators.

2.3 User Characteristics

2.3.1 Students

Users who access their grades, view courses, appeal grades, and interact with the chatbot using their digital ID cards.

2.3.2 Teachers

Users who input student marks, manage course details, deliberate on appeals, and receive assistance from the chatbot.

2.3.3 Administrators

Users who oversee the platform's overall functioning, manage accounts, and ensure data security.

2.4 Constraints

-Data privacy and security regulations (e.g., GDPR compliance).
-System must support high availability and load balancing to handle multiple users simultaneously.

2.5 Assumptions and Dependencies

-The institution provides the necessary grading rules and policies.
-Availability of a stable internet connection for accessing the platform.
-The system depends on external APIs for chatbot functionality and integration with other services if necessary.

CHAPTER 3

SPECIFIC REQUIREMENTS

3.1 Functional Requirements

3.1.1 User Authentication

-Users must be able to register and log in securely using their digital ID cards.
-Role-based access controls will differentiate students, teachers, and administrators.

3.1.2 Course Management

-Teachers can create, update, and delete courses.
-Students can view their enrolled courses.

3.1.3 Grade Management

-Teachers can input grades for assignments and exams.
-The system calculates final grades based on institution-specific rules.
-Students can view their grades, receive updates, and appeal grades if necessary.

3.1.4 Appeal and Deliberation Management

-Students can submit grade appeals through the platform.
-Teachers can review and deliberate on appeals, with the system recording outcomes.

3.1.5 Chatbot Functionality

-The chatbot will provide automated responses for frequently asked questions about courses, schedules, grades, and appeal processes.
-The chatbot will assist teachers in entering grades and other administrative tasks.

3.1.6 Digital ID Card Integration

The platform will support digital ID cards for students to authenticate their identity and access services.

3.1.7 Reporting Module

-Administrators can generate reports on student performance, appeals, and other relevant data.
-Students can download their grade reports.

3.2 Non-Functional Requirements

3.2.1 Performance Requirements

-The platform must handle at least 1,000 concurrent users.
-The chatbot should respond within 2 seconds to user queries.

3.2.2 Security Requirements

-Data encryption for all sensitive information (e.g., grades, user credentials).
-Regular security audits to ensure data protection and system integrity.
-Secure digital ID card management and authentication protocols.

3.2.3 Usability Requirements

-The platform should have a user-friendly interface accessible from various devices (PCs, tablets, smartphones).
-Accessibility compliance to accommodate users with disabilities.

3.2.4 Reliability Requirements

The platform should maintain 99.9A backup system should ensure data recovery in case of failure.

CHAPTER 4 EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

A dashboard for students, teachers, and administrators with role-specific features, including digital ID card support.

4.2 Hardware Interfaces

The platform will be accessible via standard web browsers on any compatible device.

4.3 Software Interfaces

-Integration with databases for data storage (e.g., MySQL, PostgreSQL).
-API integration for the chatbot service and ID card authentication.

4.4 Communication Interfaces

The platform will use HTTPS protocol for secure communication.

CHAPTER 5

SYSTEM FEATURES

5.1 User Dashboard

An interface for accessing courses, grades, appeals, and chatbot services using digital ID cards.

5.2 Admin Panel

A control center for administrators to manage users, courses, appeals, and system functionality.

CHAPTER 6

OTHER REQUIREMENTS

6.1 Legal and Regulatory Requirements

The platform must comply with data privacy regulations (e.g., GDPR) to protect user information.

6.2 Documentation Requirements

User manuals for students, teachers, and administrators to guide platform usage.